- Power Supplies
- Bandgap Reference
- Case Study
- Antennas

Power Supply Designs

- AC to DC power supplies *110-240vac 50/60Hz*
  - Linear
  - Switch mode
- DC to DC power supplies *+5v, +14v*
  - Linear
  - Switch mode
- Bandgap reference
- Case study

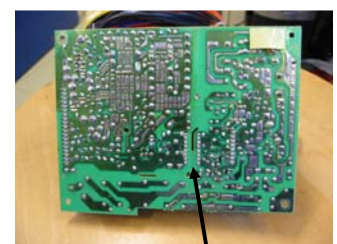
Power Supply Specifications

- Line regulation: change in output voltage with input change
- Load regulation: change in output voltage with varying load
- Output ripple
- Holdup time
- Input voltage range/frequency
- Efficiency
- Power density W/cu in
- Cost watt/$

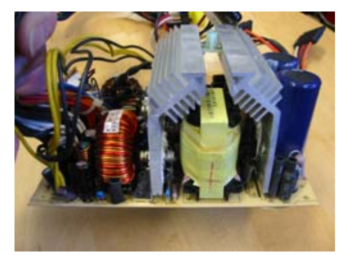
Safety

- Safety is major issue in power supplies.
- Operation world wide with one design desireable/required
- Input (primary) must be isolated from output (secondary)

DC AC line

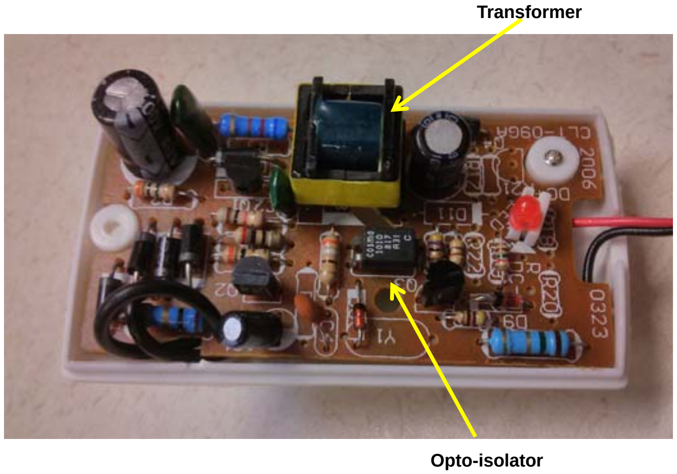


optoisolators

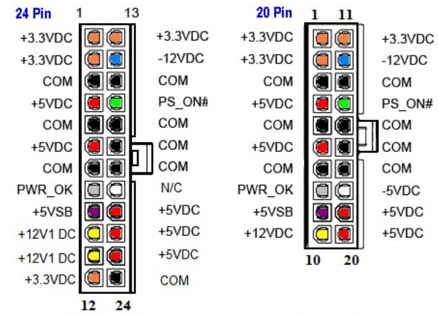


350W PC Power Supply

Primary Secondary Isolation



Connector Voltage Drop



Version 2.0 Version 1.0

ATX Motherboard power connector

- +12V currents up to 20 amps!
- Voltage drop major issue.
- Solution: additional connectors in parallel
- Must be backwards compatible

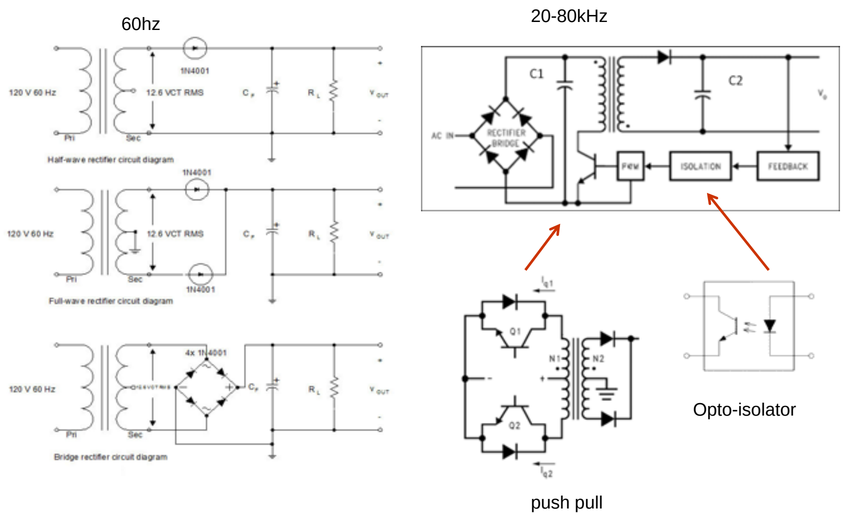
AC-DC Power Supply

- Internal: PC power supply
- Brick/Wart: USB charge, cell phone, etc…
- AC input range
  - 100-240VAC (min 92VAC in Japan)
  - 50-60Hz

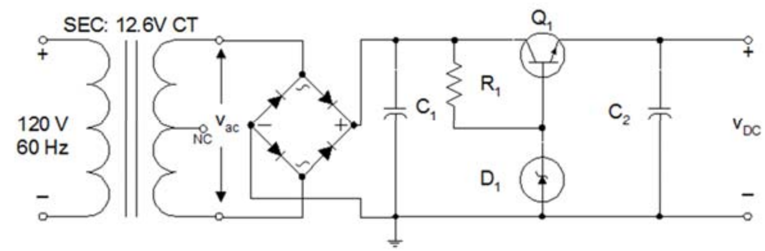
AC-DC Design Philosophy

- Step down voltage (110-240) at 50/60hz
  - Simple design
  - Large transformer required
- Off-Line Switching
  - Rectify line voltage, step down at 20 - >100kHz
  - Small transformer
  - More complex design
  - Switching noise filter required

Step Down Design Philosophy

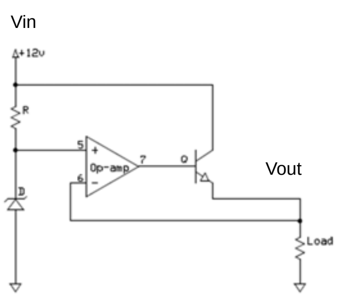


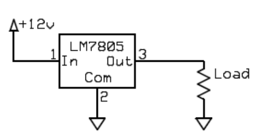
Linear Regulator: Zener + BJT



- Simple design
- Output voltage varies (slightly) with current
- Temperature drift
- BJT power dissipation limited
- Low efficiency

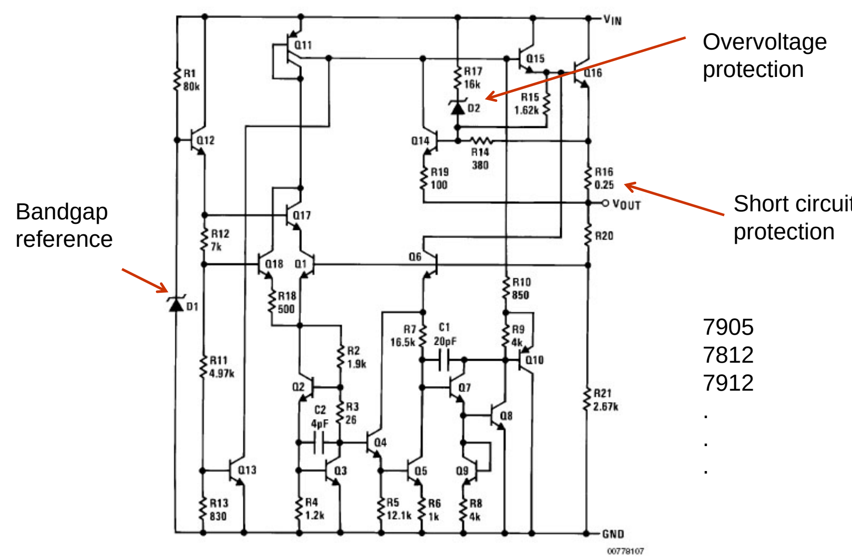
Linear Voltage Regulator





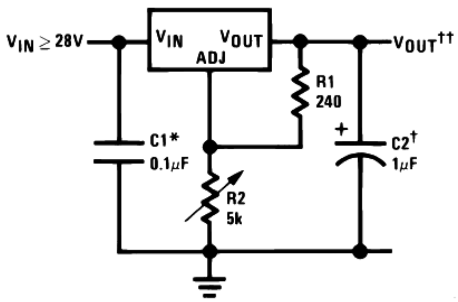
- Simple/low cost design
- Low noise/low ripple
- Fast transient response
- Low dropout voltage
- $$Efficiency = \frac{V_{out}I}{V_{out}I + (V_{in} - V_{out})I} = \frac{V_{out}}{V_{in}}$$

7805



7905
7812
7912
.
.
.

LM317

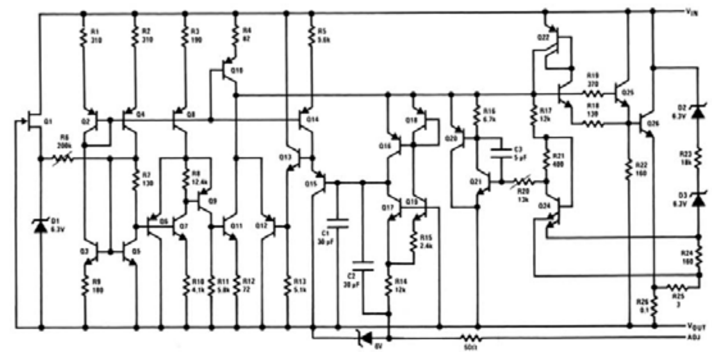


*Needed if device is more than 6 inches from filter capacitors.

†Optional — improves transient response. Output capacitors in the range of 1µF to 1000µF of aluminum or tantalum electrolytic are commonly used to provide improved output impedance and rejection of transients.

$$\dagger\dagger V_{OUT} = 1.25V\left(1 + \frac{R2}{R1}\right) + I_{ADJ}(R_2)$$

LM317 – Three Terminal Adjustable Regulator



- First 3 terminal adjustable voltage regulator
- 1.2 - 25 Voltage output range
- Short circuit protected
- Thermal shutdown

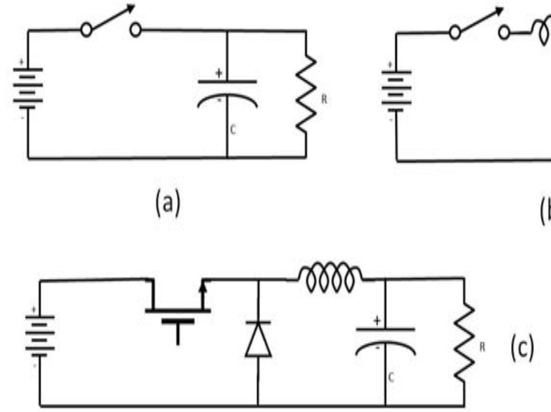
Switching Regulators

- Buck converter – step down converter
- Boost converter – step up
- Flyback converter
- Can be used to generate multiple voltages from single source.
- *Extremely efficient*

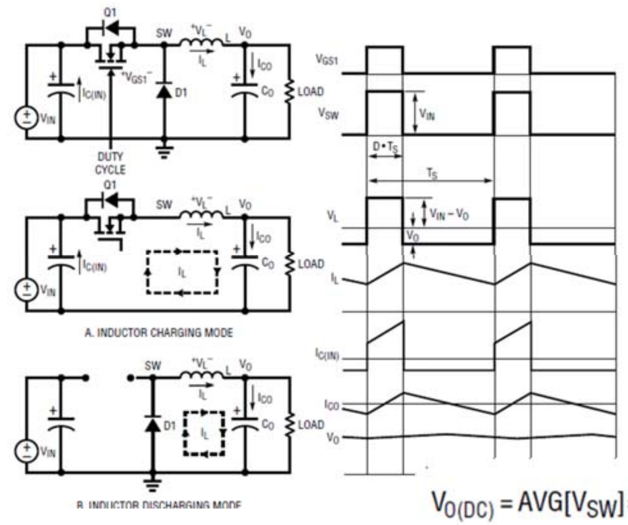
Inductor Capacitor Behavior

- Current through an inductor cannot be changed instantly.
  Since V = L(di/dt), a step change in i would imply an infinite voltage.
  - Result: The current through an inductor just before the switching equals the current just after.
- Steady state voltage across an ideal inductor must be zero.
  A steady state voltage would imply a constant, nonzero di/dt which results in infinite current.
  - Result: In equilibrium, the voltage across an ideal inductor is zero. (real inductors have resistance which will lead to an IR drop) .
- Voltage across a capacitor cannot be changed instantly. Since i=C(dv/dt) a step change in V would imply an infinite current .
  - Result: the voltage before the switching or pulse equals the voltage just after.
- Steady state current in a capacitor must be zero. A steady state current would integrate to an infinite charge and infinite voltage.
  - Result: in the steady state the average current into a capacitor is zero.

Voltage Schemes



Buck Converter* with MOSFET

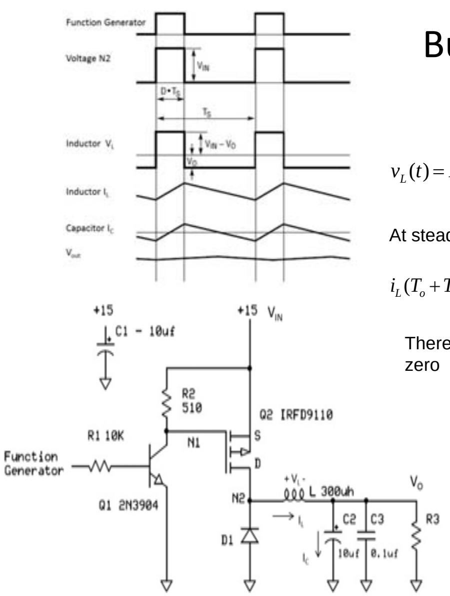


$V_{out} < V_{in}$

$$V_{O(DC)} = AVG[V_{SW}] = \frac{T_{ON}}{T_S} \bullet V_{IN}$$

* Linear.com Appnote AN140-1

Buck Converter



$$v_L(t) = L\frac{di_L(t)}{dt} \qquad i_L(t) = i_L(T_o) + \frac{1}{L}\int_{T_o}^{T_s} v_L(t)\,dt$$

At steady state, the current are the same at every Ts or

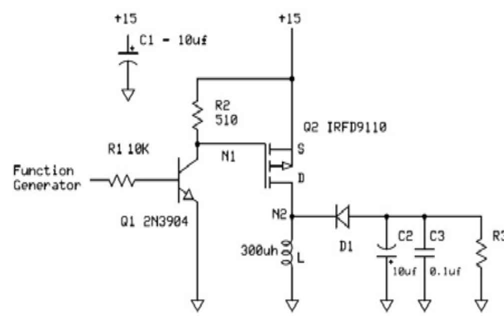
$$i_L(T_o + T_S) = i_L(T_o) \quad or \quad i_L(T_o + T_S) - i_L(T_o) = 0 = \frac{1}{L}\int_{T_o}^{T_s} v_L(t)\,dt$$

Therefore: average voltage across an inductor must be zero

$$v_{L(average)} = T_{ON} \bullet (v_{IN} - v_O) + (T_S - T_{ON}) \bullet (-v_O) = 0$$

$$\therefore \; v_O = \frac{T_{ON}}{T_S} v_{IN} \quad D = \frac{T_{ON}}{T_S} \; (duty\,cycle)$$

Inverting Converter



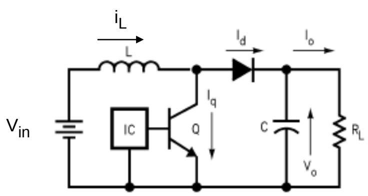
- The current following into the inductor when the MOSFET is on is: $i_L\;(on) = \frac{v_{in}}{L} T_{ON}$
- When the MOSFET is off, the diode is conducting; the change in inductor current is

  $$i_L\;(off) = \frac{v_O}{L}(T_S - T_{ON})$$

- In equilibrium, they are equal and opposite

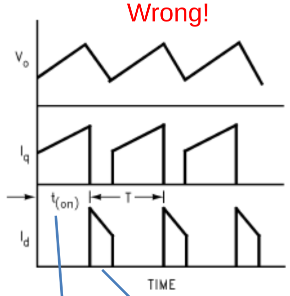
$$\therefore \; v_O = -\frac{T_{ON}}{T_S - T_{ON}} v_{IN} = \frac{T_{ON}}{T_{OFF}} v_{IN} \quad with\; T_{OFF} = T_S - T_{ON}$$

Boost Converter* with BJT



$V_o > V_{in}$

Wrong!



$$\frac{di_L}{dt} = \frac{V_{in}}{L} \qquad \frac{di_L}{dt} = \frac{V_{in} - V_o}{L}$$

Q is on for $t_{(on)}$ Q is off for $[T-t_{(on)}]$

*Texas Instruments App note AN-556 Introduction to Power Supplies

$$V_o = T * \frac{V_{in}}{(T - t_{(on)})}$$

Continuous Conduction Mode

- For light loads or low switch frequencies, the current in the inductor can fall to zero.
  - MOSFET and diode become capacitive forming a RLC circuit
- Requires more detailed analysis and design in the feedback and regulation loop.

$$f > (1-D)\left(\frac{v_O}{2 i_O L}\right)$$

$$D = \left(\frac{T_{ON}}{T_S}\right)$$

Integrated Circuit Solutions

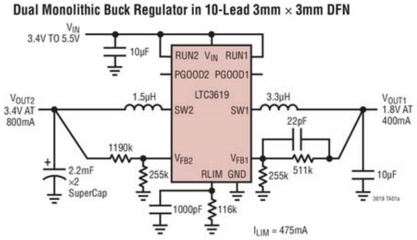
LTC3619 - 400mA/800mA Synchronous Step-Down DC/DC with Average Input Current Limit

Features

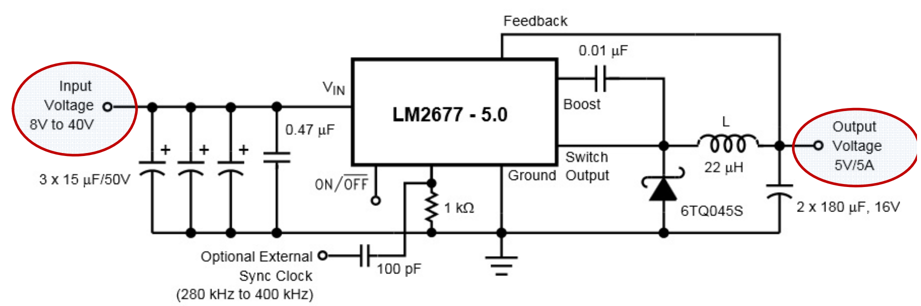
- Programmable Average Input Current Limit: ±5% Accuracy
- Dual Step-Down Outputs: Up to 96% Efficiency
- Low Ripple (<25mV$_{P-P}$) Burst Mode® Operation: I_Q = 50µA
- Input Voltage Range: 2.5V to 5.5V
- Output Voltage Range: 0.6V to 5V
- 2.25MHz Constant-Frequency Operation
- Power Good Output Voltage Monitor for Each Channel
- Low Dropout Operation: 100% Duty Cycle
- Independent Internal Soft-Start for Each Channel
- Current Mode Operation for Excellent Line and Load Transient Respo
- ±2% Output Voltage Accuracy
- Short-Circuit Protected
- Shutdown Current ≤ 1µA
- Available in Small Thermally Enhanced 10-Lead MSE and 3mm × 3mm DFN Packages

Typical Application

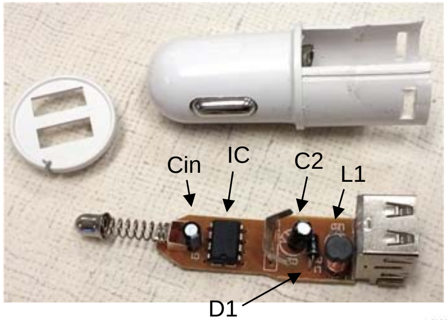
Dual Monolithic Buck Regulator in 10-Lead 3mm × 3mm DFN

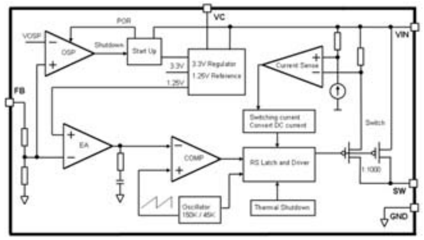


Integrated Solutions

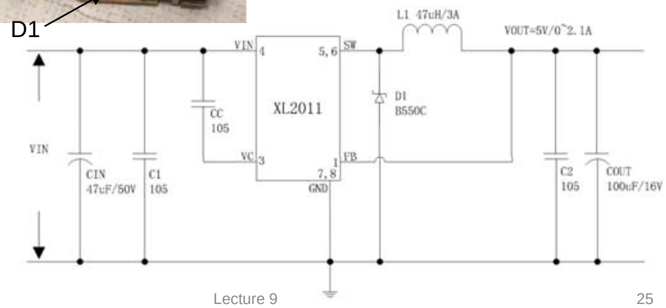


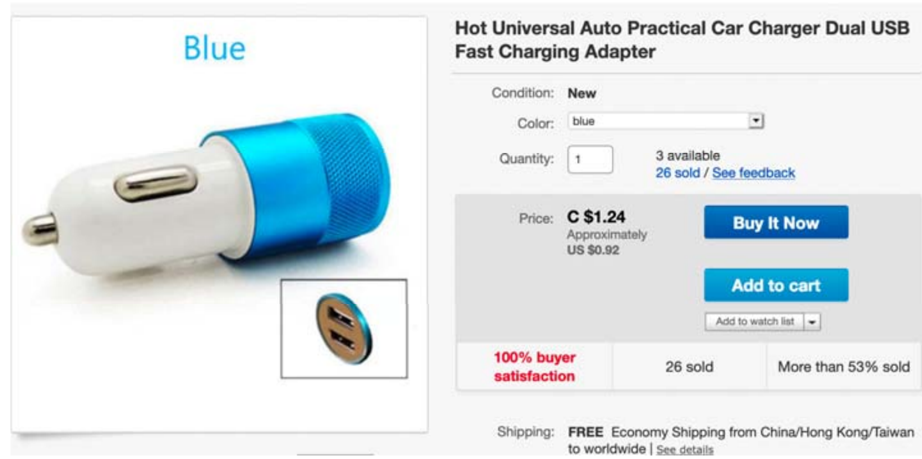
Auto USB Charger





Capacitor
104 = 10x10^4 pf
= 0.1 µf





XLSEMI Datasheet

5V/2.1A 150KHz 45V Buck DC to DC Converter For USB Interface XL2011

Features

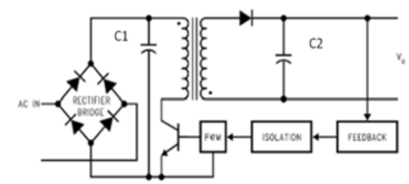
- Wide 8V to 45V input voltage range.
- Fixed 5V output voltage.
- Maximum 2.1A output current.
- Fixed 150KHz switching frequency.
- Internal optimize power MOSFET.
- High efficiency up to 92%.
- Built in output short shutdown function.
- Excellent line and load regulation.
- Built in thermal shutdown function.
- Built in current limit function.

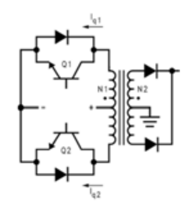
General Description

The XL2011 is a 150 KHz fixed frequency PWM buck (step-down) DC/DC converter, capable of driving a 2.1A load with high efficiency, low ripple and excellent line and load regulation. Requiring a minimum number of external components, the regulator is simple to use and include internal frequency compensation and a fixed-frequency oscillator.

Flyback Converter

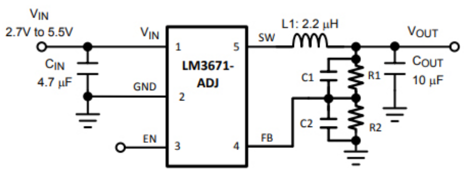
- Typically used in off-line switching regulator
- Single or push pull transistor configuration
- Transformer size approximately inversely proportional to frequency.
- Multiple output voltages possible.
- Isolation between primary and secondary absolutely essential.
- EMI line filtering necessary

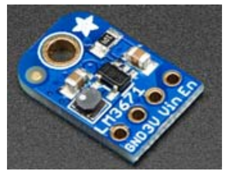




5V - 3.3V Converters

- 5VDC – existing standard for digital logic and USB
- 3.3V new standard embedded logic
  - Linear regular: LDO LM1117
  - Buck converter LM3671

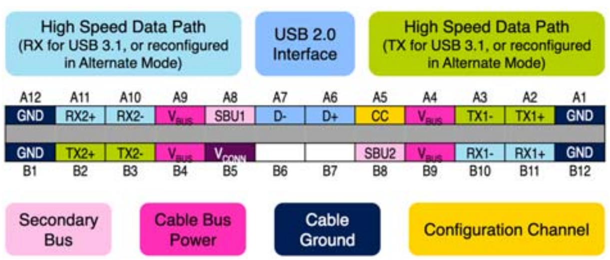


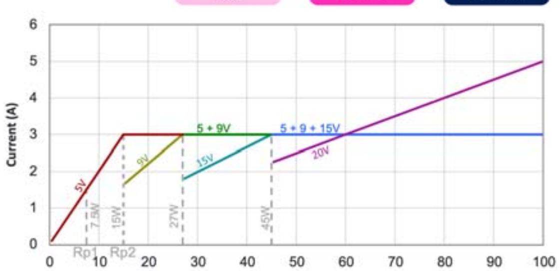


USB C

- Universal connector for power and data – first product MacBook Air – one and only port!
- Symmetrical – no orientation (Good for 10,000 insert/withdrawals)
- Supports DisplayPort, HDMI, power, USB, and VGA. Uses differential bidirectional serial communications
- Supplies up to 100W power
- Voltage dictated by software handshake, etc..
- Downside: new adapters required for DisplayPort, HDMI, power, USB, and VGA …….

USB C - Power Profiles



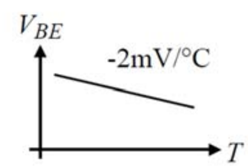


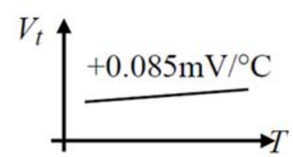
www.st.com/content/ccc/resource/sales_and_marketing/presentation/product_presentation/group0/5a/b1/8e/6c/2b/0d/46/3c/Apec/files/APEC_2016_USB_Power.pdf/_jcr_content/translations/en.APEC_2016_USB_Power.pdf

(Source: Figure 10-2 in the USB PD Specification v3.0)

Band Gap Reference

- Conceptualize by David Hibiber 1964
- Realized/implemented by Bob Widlar 1971
- Summed voltage = 1.25 (silicon bandgap voltage)*



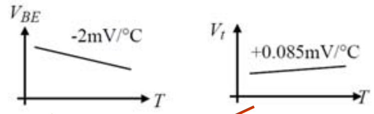


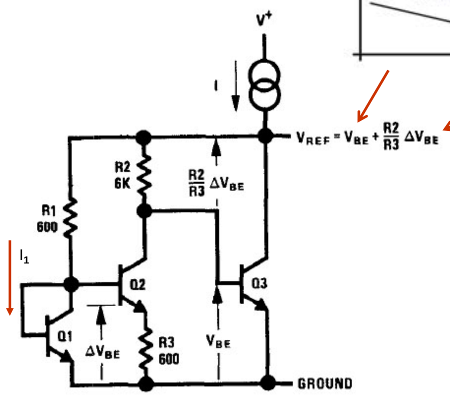
$$V_t = \frac{kT}{q}$$

$$temp\ coefficent \sim 0\ with\ V_{BE} + kV_t$$

*bandgap: amount of energy needed to free an electron from its orbit to become a mobile charge carrier.

LM309 Bandgap





$$I_B = I_S\left(e^{\left[\frac{qV_{BE1}}{kT}\right]} - 1\right) \approx I_S e^{\left[\frac{qV_{BE1}}{kT}\right]}$$

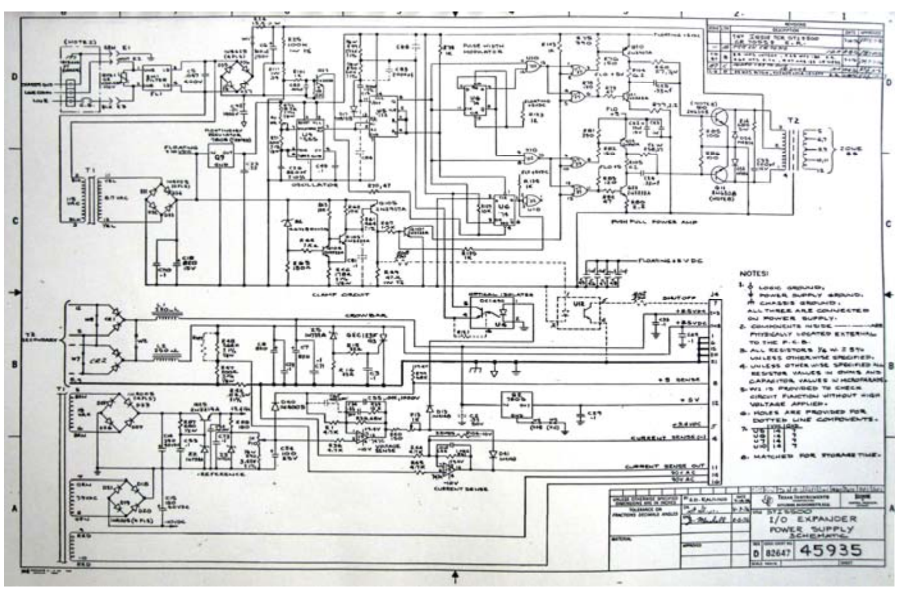
$$V_{BE1} = \frac{kT}{q}\ln(I_1) \qquad I_1 = \frac{I_B}{I_S}$$

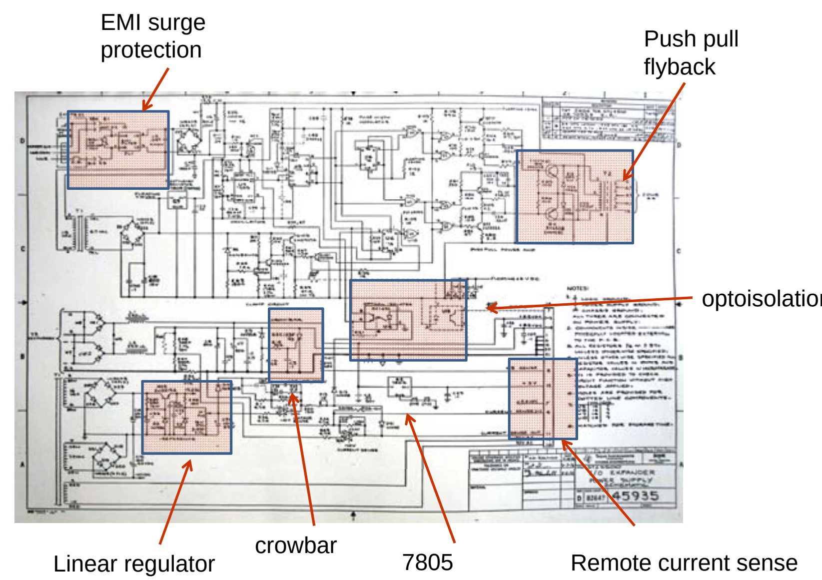
$$\Delta V_{BE} = V_{BE1} - V_{BE2} = \frac{kT}{q}\ln\left(\frac{I_1}{I_2}\right)$$

Widlar, Robert J. (February 1971), "New Developments in IC Voltage Regualtors", IEEE Journal of Solid-State Circuits 6 (1): 2–7, doi:10.1109/JSSC.1971.1050151

Case Study TI-5500 I/O Expander

- Power supply for control system
- Input: 90-120VAC 50-60Hz
- Output
  - (2) 8.5V 15 amp
  - 5V 1 amp
  - Undervoltage sense
  - Overvoltage shutdown
  - Overcurrent protection





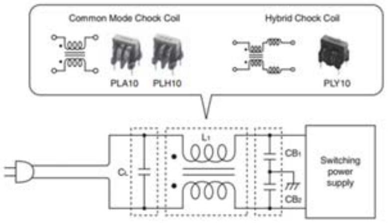
Walk Through

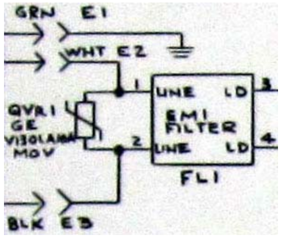
- Heat sink
- EMI
- MOV
- Opto-isolation
- Crowbar
- Fast recovery diodes
- Low ESR capacitors
- Hi Pot

Black Heatsinks

- Property: *emissivity*
  Highly polished (shiny, white) objects have low emissivity, black object have high emissivity.
- Black surfaces absorb heat better, but it also radiates it better.
- The SR-71 Blackbird is black for radiative heat transfer – cooler than unpainted. At Mach 3.5 air is heating up the plane.

EMI Filter





Symbol	Name	Function
L_1	Common mode choke coil	Suppresses common mode noise
C_L	Across the line capacitor (X-capacitor)	Suppresses differential mode noise
CB_1 CB_2	Line bypass capacitor (Y-capacitor)	Suppresses common mode noise and differential mode noise

http://www.murata.com/products/catalog/pdf/c35e.pdf

MOV

- Metal Oxide Varistor
- Zinc Oxide + other metal oxide forming small multiple back to back diodes
- Specs:
  - energy rating in joules,
  - operating voltage,
  - response time,
  - maximum current,
  - breakdown (clamping) voltage.
- Key component in surge protectors

Littlefuse MOV

Absolute Maximum Ratings

• For ratings of individual members of a series, see Device Ratings and Specifications chart

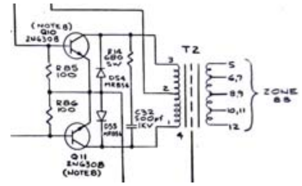
Continuous	TMOV® and iTMOV® Series	Units
Steady State Applied Voltage:		
AC Voltage Range ($V_{M(AC)RMS}$)	115 to 750	V
Transient:		
Peak Pulse Current (I_{TM}) - For 8x20µs Current Wave, single pulse	6,000 to 10,000	A
Single-Pulse Energy Capability - For 2ms Current Wave	35 to 480	J
Operating Ambient Temperature Range (T_A)	-55 to +85	°C
Storage Temperature Range (T_{STG})	-55 to +125	°C
Temperature Coefficient (αV) of Clamping Voltage (V_C) at Specified Test Current	<0.01	%/°C
Hi-Pot Encapsulation (COATING Isolation Voltage Capability)	2,500	V
Thermal Protection Isolation Voltage Capability (when operated)	600	V
COATING Insulation Resistance	1,000	MΩ
Indicator Lead Rating (Lead-3 - iTMOV® varistor only):		
Continuous RMS current	100	mA
Surge Current, 8/20µs	10,000	A

CAUTION: Stresses above those listed in "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied.

TMOV and iTMOV® Ratings & Specifications

TMOV Lead-free And RoHS Compliant Models		iTMOV Lead-free and RoHS Compliant Models		Disc Diameter	Maximum Rating (85°C)					Specifications (25°C)				
					Continuous		Transient			Varistor Voltage at 1mA Test Current		Maximum Clamping Voltage 8/20µs		Typical Capacitance f = 1MHz
					AC Volts	DC Volts	Energy 2ms	Peak Surge Current 8/20µs						
Part Number	Branding	Part Number	Branding		$V_{M(AC)RMS}$	$V_{M(DC)}$	W_{TM}	I_{TM} 1 x Pulse	I_{TM} 2 x Pulse	V_{NOM} Min	V_{NOM} Max	V_C	I_{PK}	C
				(mm)	(V)	(V)	(J)	(A)	(A)	(V)		(V)	(A)	(pF)
TMOV14RP115E	P4T115E	TMOV14RP115M	P4T115M	14	115	150	35	6000	4500	162	198	300	50	1100
TMOV20RP115E	P2T115E	TMOV20RP115M	P2T115M	20	115	150	52	10000	6500	162	198	300	100	2400
TMOV14RP130E	P4T130E	TMOV14RP130M	P4T130M	14	130	170	50	6000	4500	184	226	340	50	1000
TMOV20RP130E	P2T130E	TMOV20RP130M	P2T130M	20	130	170	100	10000	6500	184	226	340	100	1900
TMOV14RP140E	P4T140E	TMOV14RP140M	P4T140M	14	140	180	55	6000	4500	200	240	360	50	900

Push Pull Flyback



Description:

High voltage, TO-3, NPN, Silicon, Power Transistor. Designed for high voltage inverters, switching regulators and line – operated amplifier applications. Especially well suited for switching power supply applications in associated consumer products.

Features:

- Low Collector Emitter Saturation Voltage : $V_{CE(sat)}$ 1.5V(Max.) @ I_C - 3A
- Current Gain-bandwidth Product : 5MHz (Min.) @ I_C - 0.3A

Absolute Maximum Ratings:

Collector-Base Voltage, V_{CBO} : 700V
Collector-Emitter Voltage, V_{CEO} : 350V
Emitter-Base Voltage, V_{EBO} : 8V
Continuous Collector Current, I_C : 8A
Base Current I_B : 4A
Total Device Dissipation (T_C = +25°C), P_D : 125W
Derate above 25°C : 0.714mW/°C
Operating Junction Temperature Range, T_J : -65°C to +200°C
Storage Temperature Range, T_{stg} : -65°C to +200°C

Electrical Characteristics: (T_A = +25°C unless otherwise specified)

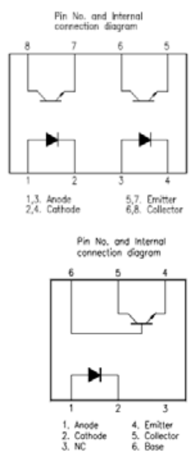
Parameter	Symbol	Test Conditions	Min.	Max.	Unit
OFF Characteristics					
Collector-Emitter Breakdown Voltage	$V_{(BR)CEO}$	Ic = 100mA, I_B = 0	350	-	V
Collector Cut-Off Current	I_{CEX}	V_{CE} = 700V, $V_{EB(off)}$ = 1.5V	-	0.5	mA
	I_{CEO}	V_{CB} = 350V, I_B = 0	-	0.5	
Emitter Cut-Off Current	I_{EBO}	V_{EB} = 8V, I_C = 0	-	1	
ON Characteristics (Note 1)					
DC Current Gain	h_{FE}	V_{CE} = 5V, I_C = 3A	12	60	-
		V_{CE} = 5V, I_C = 8A	3	-	-
Collector-Emitter Saturation Voltage	$V_{CE(sat)}$	I_C = 3A, I_B = 0.6A	-	1.5	V
		I_C = 8A, I_B = 2.67A	-	5	
Base- Emitter Saturation Voltage	$V_{BE(sat)}$	I_C = 8A, I_B = 2.67A	-	2.5	
Base-Emitter On Voltage	$V_{BE(on)}$	I_C = 3A, V_{CE} = 5V	-	1.5	

High breakdown voltage

Low beta

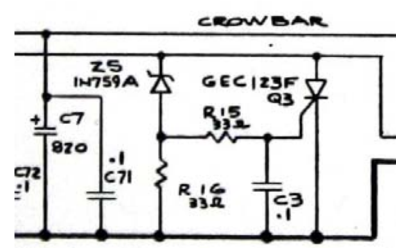
Optoisolators

- Electrically isolate circuits in two voltage domains
- Isolator achieved through vacuum or air gap
- Typical isolation: 5000 volts rms

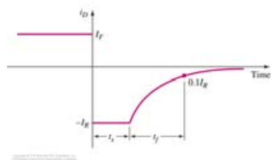


Crowbar

- Circuit to protect against overvoltage failure
- Overvoltage triggers SCR/TRIAC
- Relies on overcurrent protection or fuse.



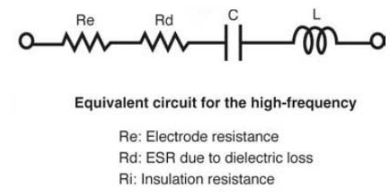
Fast Recovery Diodes



Fast recovery times achieved by manipulating doping levels and junction geometry

Description	General Purpose Rectifier	Fast Switching Rectifier	Small Signal Diode	Schottky Diode
Sample Device	1N4001	1N4933	1N4148	ZC2800
Maximum DC/Average Forward Current	1 A	1 A	300 mA	15 mA
Maximum Reverse Voltage	50 V	50 V	75 V	70 V
Reverse Leakage Current @ 25 °C and VR = 20 V	50 nA	200 nA	5 nA	200 nA
Forward Voltage	~0.7 V	1 V @ IF = 1A	1 V @ IF = 10 mA	0.41 V @ 1 mA
Reverse Recovery Time	2 µs	200 ns	4 ns	1 ns
Cost	20 cent	15 cent	25 cent	100 cent

Low ESR Capacitors



- ESR – Equivalent Series Resistance
- Electrolytic 10uf: 0.1-3Ω
- Ceramic, low ESR: <0.015Ω

Hi Pot

- Safety test to verify isolation between primary and secondary.
- High potential test

Switching Power Supply Losses

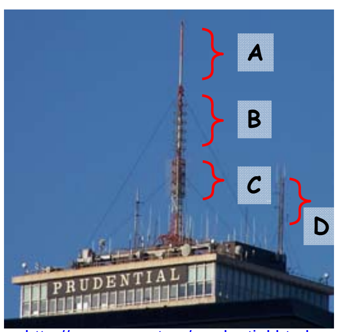
- Inductor loss
- Capacitor ESR loss
- Diode loss
- BJT/MOSFET conduction loss
- BJT/MOSFET rise/fall time loss
- Gate drive loss

Antennas

Goals

- To show the possibility of harvesting energy from the environment, and in the case of our lab from the radio station antennas on top of the Prudential center.
- To gain basic understanding of how antennas work and how to use the Friis transmission equation.
- To make an interesting analog circuit that uses the harvested energy to perform a task.

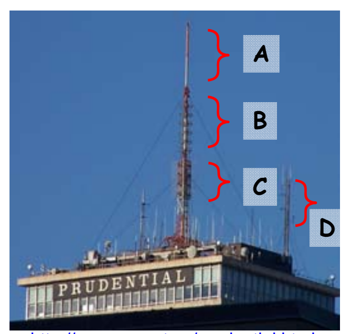
Overview of Prudential Center Antennas



http://www.necrat.us/prudential.html https://www.fybush.com/sites/2004/site-040709.html

The first TV broadcasting antennas were mounted on the Prudential tower in October 1964. FM radio antennas came a few years later.

Overview of Prudential Radios



http://www.necrat.us/prudential.html

Antenna	Frequency (MHz)	Power (kW)	Center of radiation (m)
B	100.7	21.5	235
	1004.1	21.0	
	106.7	21.5	
	107.9	20.5	
C	92.90	18.5	224
	105.7	23.0	
	96.90	22.5	

D: backup antenna

Stations
100.7: WZLX Classic Rock
105.7: WROR Classic Hits

How Far Are We from the Tower?

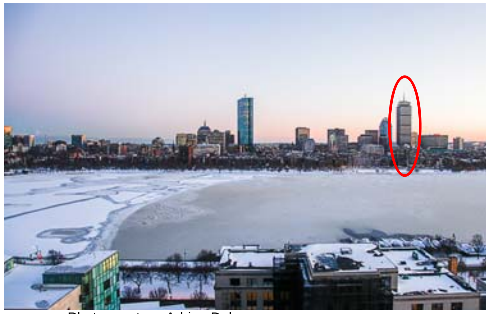
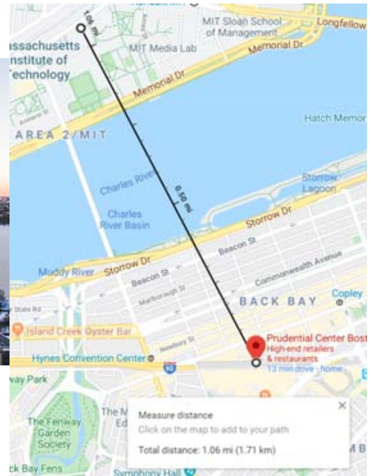


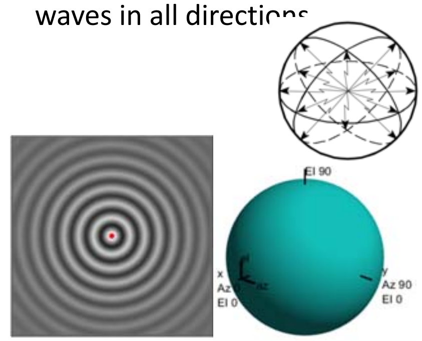
Photo courtesy Adrian Dalca



Source: Google Maps

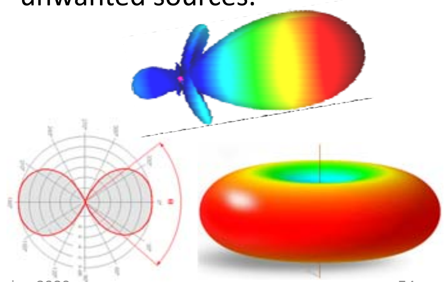
Antenna Basics

Isotropic Antenna	Directional Antenna
An isotropic antenna is a hypothetical antenna radiating the same intensity of radio waves in all directions.	An antenna which radiates or receives greater power in specific directions allowing increased performance and reduced interference from unwanted sources.



Source: Wikipedia

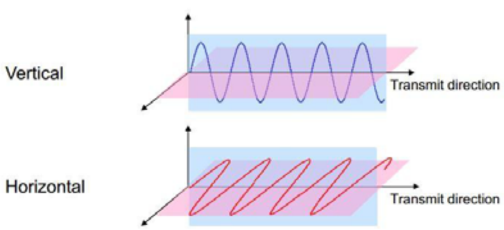
Source: Mathworks



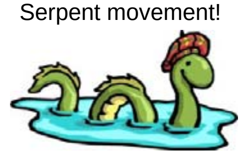
Source: https://www.radartutorial.eu/

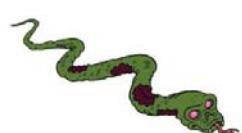
Antenna Polarization

- Electric field can be either vertical or horizontal (or a combination of both).



Serpent movement!



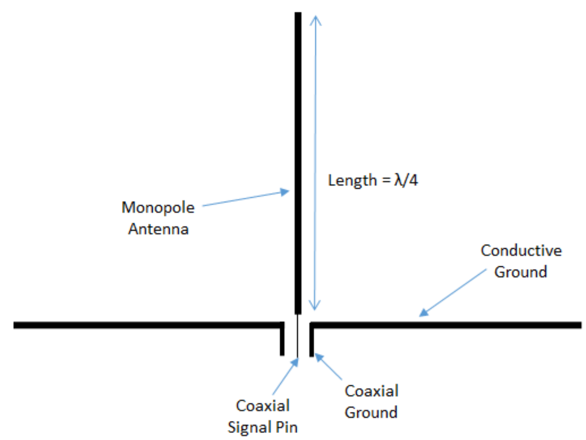


Snake movement!

Source: Google Photos

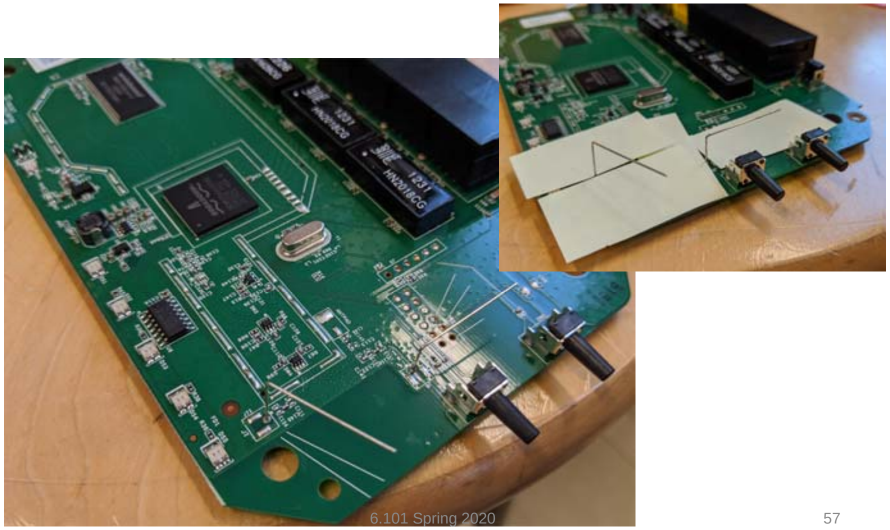
Monopole Antenna

Source: arcantenna.com



Source: microwavetools.com

Example WiFi Router Antennas



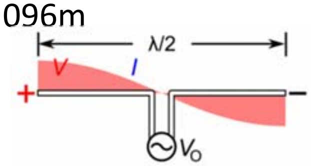
Antenna Directivity

- Antenna Gain (G) describes how much power is transmitted in the direction of radiation to that of an isotropic source.
- Usually reported in dBi unit.

$$G_{dBi} = 10\log_{10}(G)$$

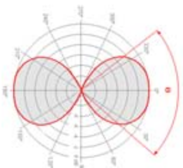
Half-wave (Dipole) Antenna

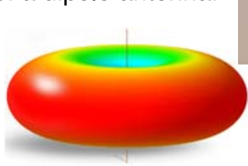
- For the 96.9MHz channel, λ is = 3.096m
- So $\frac{\lambda}{2} = \frac{c}{2f}$ =154.8cm



- Directivity Gain: 2.15dBi (G =1.64)

Radiation Pattern of a dipole antenna:

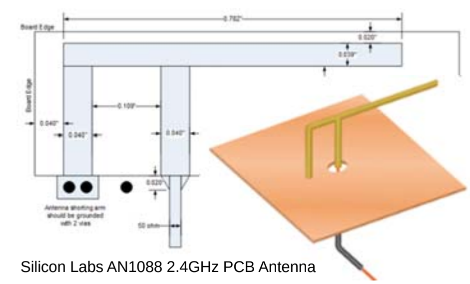




Other Types of Antennas

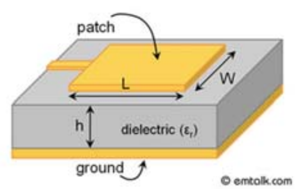
Horn Antenna

Inverted-F Antenna

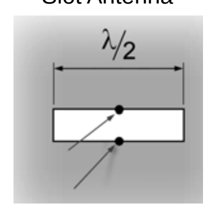


Silicon Labs AN1088 2.4GHz PCB Antenna

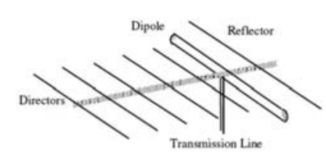
Patch Antenna



Slot Antenna



Yagi-Uda Antenna



Source: Google Photos

Prudential Tower Antenna Gain

- Let's assume the tower is equipped with an isotropic hemisphere antenna.
- Meaning that the radiated power is isotopically distributed in half the sphere.
- This means

$$G_{dBi} = 10\log_{10}(2) = 3\text{dBi}$$

Friis Transmission Equation

- The "Friis Transmission Equation" is used to calculate the power received from one antenna (with gain G_r), when transmitted from another antenna (with gain G_t), separated by a distance r, and operating at wavelength λ.

$$P_r = G_t G_r P_t (\frac{\lambda}{4\pi r})^2$$

Power we can receive at MIT?

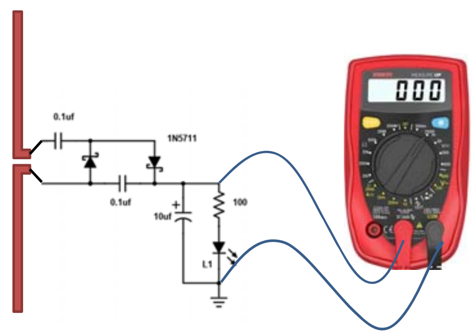
- For a transmitted power of 22.5kW (see above), we can receive:

$$P_r = G_t G_r P_t (\frac{\lambda}{4\pi r})^2$$

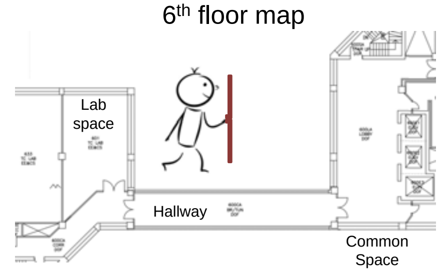
$$P_r = 2 \times 1.64 \times 22.5 \times 10^3 \times (\frac{3.096}{4\pi \times 1700})^2 \simeq 1.55\ mW$$

Prudential Center

Find the Optimal Spot to Maximize Your Reading



6th floor map



Answer the following questions:

What polarization would maximize your reading?

What location would maximize your reading?

What is the maximum voltage that you can read?

What Else Can We Do?

- With this power harvester design an interesting circuit:
  - Flashing light
  - Generating a beep
  - Activating a buzzer
- Please note that the higher the power required, the bigger the capacitor and the longer the charging time for the capacitor.